HARDWARE TRACK

1. **Design a Miniature Satellite Health Monitor**
   Design a miniature satellite subsystem that monitors its internal temperature, power level, and operational status. It must transmit telemetry to a ground station and alert the operator when an abnormal condition occurs.

2. **Intelligent Power Fault Detection**
   Design a monitoring unit that detects abnormal voltage, excessive current, and unexpected power consumption in a small electronic system, then reports the fault and takes appropriate protective action.

3. **IoT Communication Failure Recovery**
   Build an IoT system that continues recording critical sensor data and recovers gracefully when Wi-Fi or the central server becomes unavailable.

4. **Sensor-to-Actuator Automation System**
   Develop a compact sorting system that identifies objects using a sensor or camera and routes them to the correct destination using a physical actuator.

5. **Multi-Rover Coordination Without a Central Controller**
   Design a system in which three simulated or real autonomous rovers divide exploration or delivery tasks, avoid conflicting routes, and continue operating when one rover becomes unavailable.
   NOTE: ROS2 simulations are allowed, Basics RC cars are not recommended.

6. **Autonomous Rover Docking and Recovery**
   Develop a rover that autonomously approaches a docking station, aligns with a target, confirms successful docking, and attempts recovery if the initial approach fails.
   NOTE: ROS2 simulations are allowed, Basics RC cars are not recommended.

7. **Edge-AI PCB Inspection & Automated Defect Localization**
   Develop a low-cost, edge-AI-powered system to inspect Printed Circuit Boards (PCBs), detect visible manufacturing defects, and localize them using computer vision. Participants can integrate reference-image comparison, defect classification, and board-coordinate mapping and also a solution to correct the design either by a system (for eg. Robotic arm) or Schematics in software.

SOFTWARE TRACK

1. **Real-Time Cyberattack Detection**
   Build an intelligent security monitoring platform that analyzes server logs, authentication events, and network activity to detect multi-stage cyberattacks.

2. **AI-Powered Anomaly & Rare Celestial Event Discovery Platform**
   Create an AI-powered discovery platform that analyzes astronomical observations and identifies unusual celestial objects or rare events hidden in large scientific datasets.

3. **Dynamic Transit & Emergency Re-Routing System**
   Design a dynamic routing engine that finds efficient routes for emergency vehicles or public transport when roads become congested, blocked, or unavailable. The system should update route paths in real time as roadblocks occur, clear priority paths for ambulances, and provide live guidance to keep response times fast.

4. **Privacy-Preserving Eligibility Verification Platform**
   Digital services often demand more personal information than is necessary to verify a user's eligibility. For example, a service may need to confirm that someone is above a minimum age or meets a financial threshold but should not need their exact birth date or account balance. Develop a privacy-preserving verification platform that allows a user to prove that a condition is satisfied without revealing the underlying sensitive information.

5. **Automated Fact Checking & Misinformation Evolution Tracker**
   Build a claim-verification platform that compares a statement against a collection of trusted sources, identifies contradictory evidence, and traces how a claim changes as it spreads. The system should map how rumours mutate over time, flag false claims, and give a simple credibility score backed by trusted links.

6. **Multi-Layer Deepfake & Synthetic Media Defence**
   Create a multi-layer verification system to detect highly convincing, AI-generated deepfakes. The platform should combine visual and audio forensic analysis, biological signals (such as eye blinks or facial pulse), and cryptographic checks to reliably spot manipulated media.

7. **Offline P2P Disaster Rescue & Hazard Mapping Network**
   Engineer a decentralized disaster communication system that enables offline smartphones to form local peer-to-peer mesh networks when cellular infrastructure fails. The platform should allow stranded individuals and first responders to exchange critical hazard locations, verified safe shelters, and SOS distress alerts across nearby devices without internet access.

OPEN INNOVATION TRACK

1. Best Use of Gemma 4

- **Description**: Go beyond conventional AI applications and demonstrate how Gemma 4 can power meaningful, innovative solutions to real-world problems.
- Gemma 4 offers powerful AI capabilities, but simply integrating an AI model into an application doesn't make it innovative or valuable. We want to see projects where AI plays a fundamental role in solving a real-world problem, not just acting as a wrapper around an existing API.
- Avoid conventional, overdone ideas such as generic AI assistants, AI doctor chatbots, basic crop yield prediction systems using the Gemma API, or applications that simply send user input to a model and display its response.
- **Requirements**:
    - **Problem-first thinking:** Start with a real problem, not with the desire to use AI.
    - **Meaningful AI integration:** Gemma 4 should play a central role in the solution.
    - **Originality:** Go beyond repetitive, overused AI project ideas and explore genuinely useful applications.
    - **Working implementation:** Build a functional prototype that demonstrates how your solution addresses the problem.
- **Participant Resources:**
    - Gemma 4 resources
    - Gemma Quickstart
    - Gemma API docs
    - Gemma Beginner Guide

- Pick problems shaped by what's happening in tech right now—like running AI locally on small devices without internet (on-device / edge AI), using AI to take real actions via tools and code (AI agents), or processing real-world data like audio, sensor signals, and complex documents instead of just typing in a chat box.

  The goal is not to build something that uses AI. The goal is to build something valuable because of AI.

2. Best Open-Source AI Project

- **Description**: Build an original project where open-source or open-weight AI is a core feature. You can integrate an open-weight language model (LLM/SLM), create an agent skill, or build/adapt an open-source model harness. Feel free to use one of these approaches or combine them!

- **Requirements**:
    - **Core Feature**: Open-source or open-weight AI must be a central part of how your project functions.
    - **Public & Open:** Your code must be published in a public GitHub repository under an open-source license (e.g., MIT, Apache-2.0, GPL-3.0).
    - **Agent Skills**: If you build an agent skill, it must comply with the Agent Skill Open Standard.
    - **Model Harness:** If you submit a model harness, it must be an original build or feature meaningful, significant changes to an existing open-source harness.

- **What Do These Terms Mean?**
  - **Open source**: the project's source code is public and carries a license that allows others to use, modify, and share it. Common choices include MIT, Apache-2.0, and GPL-3.0. The OSI license list includes more options.
  - **Open-weight model**: the trained model weights are available to download or run under the model provider's terms. Open weight does not automatically mean that every part of a model is open source, so teams should name the model and link its license or terms in the README.
  - **Small and large language models**: for this challenge, a small language model has 10 billion parameters or fewer, while a large language model has more than 10 billion parameters.
  - **Agent skill**: a reusable set of instructions and supporting files that follows the Agent Skills open standard.
  - **Model harness**: the software around a model that manages prompts, tools, memory, context, and actions. Teams may build one from scratch or make meaningful improvements to an existing open-source project such as OpenCode, Hermes Agent, OpenClaw, or Pi.